

Question 1

- a) Solve $\frac{x+2}{x} \geq 3$. 3
- b) i) Graph $y = |2x - 4|$ and $y = x$ on the same number plane. 2
ii) For what values of c does the equation $|2x - 4| = x + c$ have exactly two solutions? 2
- c) Solve $|x + 1| + |x - 2| = 3$. 2
- d) i) Expand $(x - 3)^3$. 1
ii) Write down the general term T_{k+1} in the expansion of $\left(x - \frac{1}{x}\right)^{10}$. 2
iii) Hence or otherwise, find the coefficient of x^6 in the expansion of $(x - 3)^3 \left(x - \frac{1}{x}\right)^{10}$. 3

End of Question 1

Question 2 (Start a new page)

- a) Find the exact value of $\frac{\tan 15^\circ}{1 + \tan^2 15^\circ}$. 2
- b) Find possible values of a if the lines $ax + 3y - 10 = 0$ and $ax - 2y + 5 = 0$ are inclined to each other at an angle θ , where $\tan \theta = 2$. 3
- c) Using the expansion of $\sin(A + B)$, show that $\sin 15^\circ = \frac{\sqrt{6} - \sqrt{2}}{4}$. 3

Question 2 (continued from previous page)

- d) Solve $\cos 4x + \cos 2x = 0$, for $0^\circ \leq x \leq 180^\circ$ **3**
- e) i) Sketch the graph of $y = 2 \sin 3x$ for $0^\circ \leq x \leq 180^\circ$. **3**
- ii) Hence, by drawing a suitable line on this graph, find the **number of solutions** when $\sin 3x = 1$ is solved for $0^\circ \leq x \leq 180^\circ$? **1**

End of Question 2

Question 3 (Start a new page)

- a) i) Prove the identity $\frac{1}{\tan A + \cot B} + \frac{1}{\cot A + \tan B} = \frac{\sin(A+B)}{\cos(A-B)}$. **2**
- ii) Hence show that $\frac{2}{\tan A + \cot A} = \sin 2A$. **1**
- b) Using t – formula, solve $2 \sec x + \tan x = 2$, correct your answers to the nearest degree. **4**
- c) For $0^\circ \leq x \leq 360^\circ$, solve $\cos x - \sqrt{3} \sin x = -\sqrt{3}$ by first expressing it in the form of $R \cos(x - \theta)$. **3**
- d) The window of a lighthouse is observed from A, due south of the lighthouse at an angle of elevation of α . From B, which is due east of A, the same window is observed at an angle of elevation of β . **C is the midpoint of AB.**

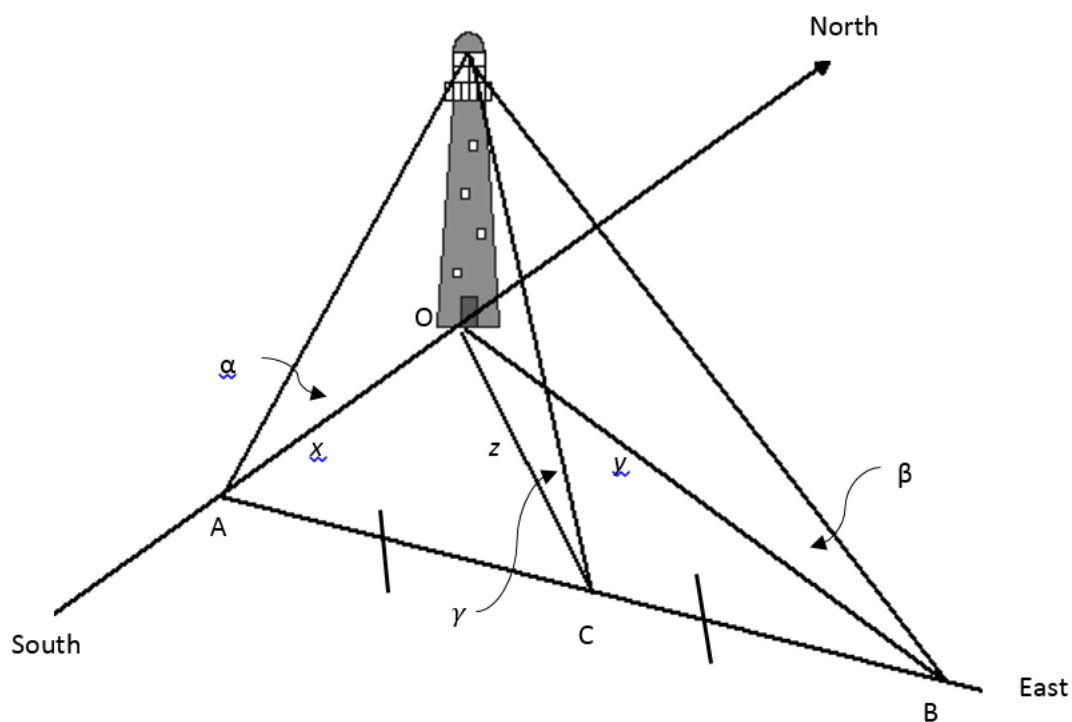
Diagram on next page

Question 3 (continued from previous page)

- i) Let $OA = x$, $OB = y$ and $OC = z$ as shown. Show that
 $x = h \cot \alpha$, $y = h \cot \beta$ and $z = h \cot \gamma$. 1

- ii) Show that $AC = \frac{\sqrt{y^2 - x^2}}{2}$. 1

- iii) Show that the angle of elevation, γ , of the window from C can be calculated
 from $\tan \gamma = \left(\frac{2}{\sqrt{3 \cot^2 \alpha + \cot^2 \beta}} \right)$. 3



End of Question 3

End of paper

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Q1

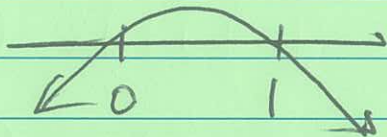
(a) $\frac{x+2}{x} \geq 3, x \neq 0$

$$x^2 \times \frac{x+2}{x} \geq 3x^2$$

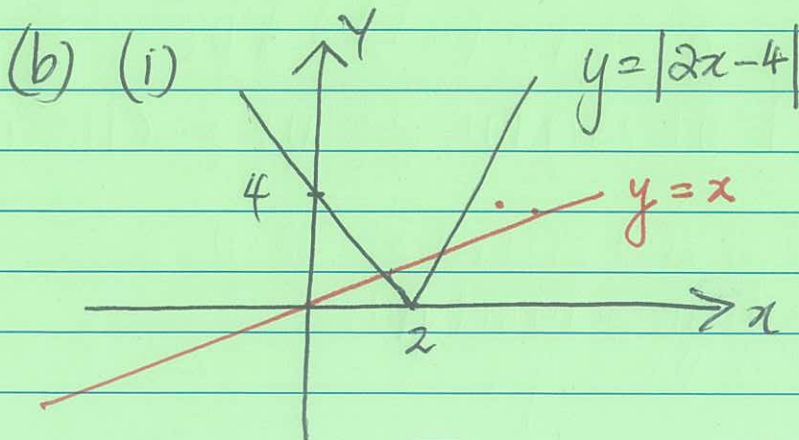
$$x(x+2) - 3x^2 \geq 0$$

$$x(2-2x) \geq 0$$

$$2x(1-x) \geq 0$$



$$0 < x \leq 1$$



(ii)

$$\begin{aligned} 2x - 4 &= x \\ x &= 4 \end{aligned}$$

$$\begin{aligned} -(2x - 4) &= x \\ 4 &= 3x \\ x &= \frac{4}{3} \end{aligned}$$

$$\therefore \text{For } |2x - 4| < x$$

$$\frac{4}{3} < x < 4$$

Q1

$$\begin{array}{rcl} \text{(1)} & x + y - z = 1 & \text{--- (1)} \\ & 8x + 3y - 6z = 1 & \text{--- (2)} \\ & -4x - y + 3z = 1 & \text{--- (3)} \end{array}$$

From (1) $z = x + y - 1$ --- (4)

Sub (4) into (2)

$$\begin{aligned} 8x + 3y - 6(x + y - 1) &= 1 \\ 8x + 3y - 6x - 6y + 6 &= 1 \\ 2x - 3y &= -5 \quad \text{--- (5)} \end{aligned}$$

Sub (4) into (3)

$$\begin{aligned} -4x - y + 3(x + y - 1) &= 1 \\ -4x - y + 3x + 3y - 3 &= 1 \\ -x + 2y &= 4 \\ x &= 2y - 4 \quad \text{--- (6)} \end{aligned}$$

Sub (6) into (5)

$$\begin{aligned} 2(2y - 4) - 3y &= -5 \\ 4y - 8 - 3y &= -5 \\ y &= -5 + 8 \\ y &= 3 \end{aligned}$$

Sub. $y = 3$ into (6)

$$x = 2(3) - 4 = 2$$

Sub. x, y into (4), $z = 2 + 3 - 1 = 4$

$$\therefore x = 2, y = 3, z = 4$$

Q1

(d)

$$(i) \quad (x-3)^3 = 1(x)^3 + 3(x)^2(-3) + 3(x)(-3)^2 + (-3)^3 \\ = x^3 - 9x^2 + 27x - 27$$

$$(ii) \quad (2x-1)^6 = \dots \dots \dots \binom{6}{4} (2x)^2 (-1)^4 \\ + \binom{6}{5} (2x)^1 (-1)^5 + \dots \dots \dots$$

In the expansion $(x-3)^3 (2x-1)^6$, the coeff. of x^2 is

$$-9 \times (-1)^4 \binom{6}{6} + 27 \times \binom{6}{5} (2) (-1)^5 - 27 \binom{6}{4} (2)^2 (-1)^4 \\ = -9 + 27(6)(2)(-1) - 27(15)(4) \\ = -1953$$

$$(e) \quad (1-(x+1))^6 = (1-x-1)^6 = (-x)^6 = x^6$$

Q2

$$(a) \quad \frac{2 \tan 15^\circ}{1 + \tan^2 15^\circ} = \sin 30^\circ = \frac{1}{2}$$

$$\therefore \frac{\tan 15^\circ}{1 + \tan^2 15^\circ} = \frac{1}{2} \left(\frac{1}{2} \right) = \frac{1}{4}$$

$$(b) \quad m_1 = \frac{a}{-3}, \quad m_2 = \frac{a}{2}$$

$$\tan \theta = 2 = \left| \frac{\frac{a}{2} - \left(-\frac{a}{3} \right)}{1 + \left(-\frac{a}{3} \right) \left(\frac{a}{2} \right)} \right|$$

$$\pm 2 = \frac{5a}{6 - a^2}$$

$$2 = \frac{5a}{6 - a^2}$$

$$12 - 2a^2 = 5a$$

$$2a^2 + 5a - 12 = 0$$

$$(2a - 3)(a + 4) = 0$$

$$a = \frac{3}{2}, a = -4$$

$$-2 = \frac{5a}{6 - a^2}$$

$$2a^2 - 5a - 12 = 0$$

$$(2a + 3)(a - 4) = 0$$

$$a = -\frac{3}{2}, a = 4$$

$$\therefore a = \pm \frac{3}{2}, \pm 4$$

Alternative approach to Q2(c)

$$\cos x - \sqrt{3} \sin x = -\sqrt{3}$$

$$= R \cos(x - \theta)$$

$$= R \cos x \cos \theta + R \sin x \sin \theta \quad \checkmark$$

Comparing terms,

$$R \cos \theta = 1 \quad \text{--- ①}$$

$$R \sin \theta = -\sqrt{3} \quad \text{--- ②}$$

$$\frac{\text{②}}{\text{①}} \Rightarrow \tan \theta = -\sqrt{3} \Rightarrow \theta = -60^\circ \quad \checkmark$$

$$\text{①}^2 + \text{②}^2 \Rightarrow R = 2.$$

$$\therefore \cos x - \sqrt{3} \sin x = -\sqrt{3}$$

$$2 \cos(x - -60^\circ) = -\sqrt{3}$$

$$\cos(x + 60^\circ) = -\frac{\sqrt{3}}{2}$$

$$x + 60^\circ = 150^\circ, 210^\circ \quad \checkmark$$

$$x = 90^\circ, 150^\circ$$

Q2

(c) $\cos x - \sqrt{3} \sin x = -\sqrt{3}$

Let $\cos x - \sqrt{3} \sin x = R \cos(x + \alpha)$

$$= R \cos x \cos \alpha - R \sin x \sin \alpha$$

$$R \cos \alpha = 1 \dots \textcircled{1}$$

$$R \sin \alpha = \sqrt{3} \dots \textcircled{2}$$

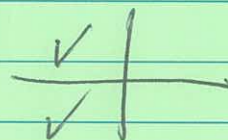
$$R = \sqrt{1 + \sqrt{3}^2} = 2$$

$$\frac{\textcircled{2}}{\textcircled{1}}, \tan \alpha = \sqrt{3}$$

$$\alpha = 60^\circ$$

$$\therefore 2 \cos(x + 60^\circ) = -\sqrt{3}$$

$$\cos(x + 60^\circ) = -\frac{\sqrt{3}}{2}$$



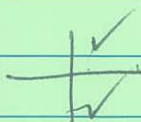
$$x + 60^\circ = 150^\circ, 210^\circ$$

$$x = 90^\circ, 150^\circ$$

(d) $\cos 4x + \cos 2x = 0$

$$2\cos^2 2x - 1 + \cos 2x = 0$$

$$(2\cos 2x - 1)(\cos 2x + 1) = 0$$



$$\cos 2x = \frac{1}{2}, \cos 2x = -1$$

$$2x = 60^\circ, 300^\circ$$

$$x = 30^\circ, 150^\circ$$

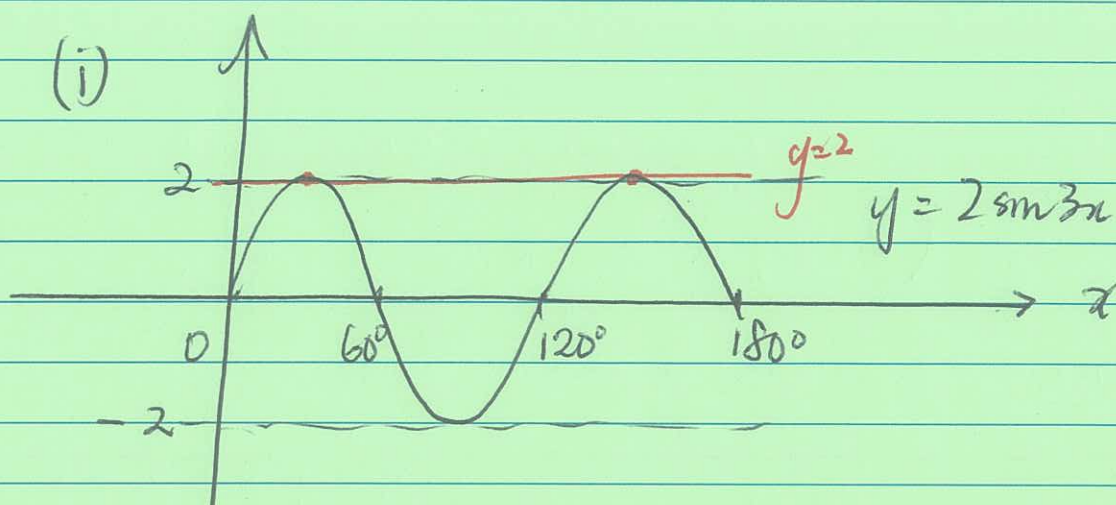
$$2x = 180^\circ$$

$$x = 90^\circ$$

$$\therefore x = \underline{30^\circ, 90^\circ, 150^\circ}$$

Q2

(e) (i)



(ii) $\sin 3x = 1$
 $2 \sin 3x = 2.$

Draw $y = 2$

2 solutions

Q3

(a) (i)

$$\text{LHS} = \frac{1}{\frac{\sin A}{\cos A} + \frac{\cos B}{\sin B}} + \frac{1}{\frac{\cos A}{\sin A} + \frac{\sin B}{\cos B}}$$

$\times \cos A \sin B$ $\times \sin A \cos B$
 $\times \cos A \sin B$ $\times \sin A \cos B$

$$= \frac{\cos A \sin B}{\sin B \sin A + \cos B \cos A} + \frac{\sin A \cos B}{\cos A \cos B + \sin A \sin B}$$

$$= \frac{\sin A \cos B + \cos A \sin B}{\sin A \sin B + \cos A \cos B}$$

$$= \frac{\sin(A+B)}{\cos(A-B)} = \text{RHS}$$

(ii) From (i), let $B = A$

$$\frac{1}{\tan A + \cot A} + \frac{1}{\cot A + \tan A} = \frac{\sin 2A}{\cos 0^\circ}$$

$$\therefore \frac{2}{\tan A + \cot A} = \sin 2A \quad \text{as } \cos 0^\circ = 1$$

Q3

$$(b) \quad 2 \sec x + \tan x = 2$$

$$2 \left(\frac{1+t^2}{1-t^2} \right) + \frac{2t}{1-t^2} = 2$$

$$\cancel{1+t^2} + t = \cancel{1-t^2}$$

$$2t^2 + t = 0$$

$$t(2t+1) = 0$$

$$t = 0$$

$$t = -\frac{1}{2}$$

$$\tan \frac{x}{2} = 0$$

$$\frac{x}{2} = 0^\circ, 180^\circ, 360^\circ$$

$$x = 0^\circ, 360^\circ$$

$$\tan \frac{x}{2} = -\frac{1}{2}$$

$$\frac{x}{2} = 180^\circ - 27^\circ, 360^\circ - 27^\circ \quad \begin{array}{c} \checkmark \\ | \\ \checkmark \end{array}$$

$$x = 306^\circ$$

Test $x = 180^\circ$,

$$\frac{2}{\cos 180^\circ} + \tan 180^\circ = \frac{2}{-1} + 0 = -2 \neq 2.$$

$\therefore 180^\circ$ is not a solution.

$$x = 0^\circ, 306^\circ, 360^\circ$$

Q3

$$(i) (1) \tan \alpha = \frac{h}{x}$$

$$x = \frac{h}{\tan \alpha} = h \cot \alpha$$

Similarly $y = h \cot \beta$
 $z = h \cot \gamma$

(ii)

$$(AC + CB)^2 + x^2 = y^2$$

$$(2AC)^2 + x^2 = y^2$$

as $AC = CB$

$$4AC^2 + x^2 = y^2$$

$$AC = \frac{\sqrt{y^2 - x^2}}{2}$$

(iii)

$$AC^2 + x^2 = z^2$$

$$\frac{y^2 - x^2}{4} + x^2 = z^2$$

$$y^2 + 3x^2 = 4z^2$$

$$h^2 \cot^2 \beta + 3h^2 \cot^2 \alpha = 4h^2 \cot^2 \gamma$$

$$4 \cot^2 \gamma = \cot^2 \beta + 3 \cot^2 \alpha$$

$$\frac{4}{\tan^2 \gamma} = \cot^2 \beta + 3 \cot^2 \alpha$$

$$\frac{4}{\cot^2 \beta + 3 \cot^2 \alpha} = \tan^2 \delta$$

$$\therefore \tan \delta = \frac{2}{\sqrt{\cot^2 \beta + 3 \cot^2 \alpha}}$$